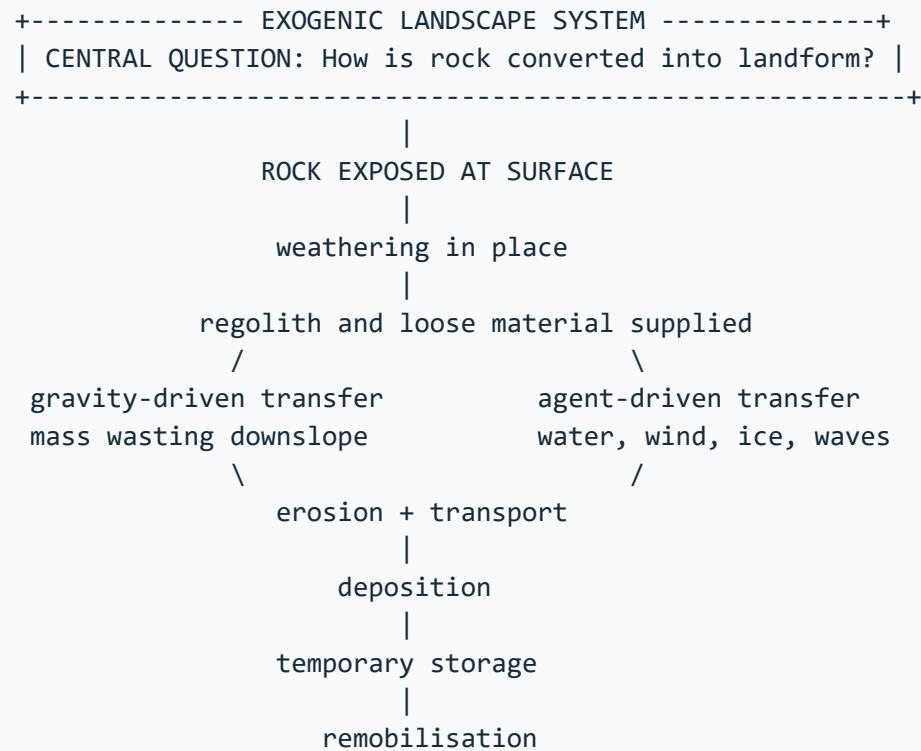


ASCII MASTER FLOW – PANEL 1/12: Exogenic source-transfer-storage system

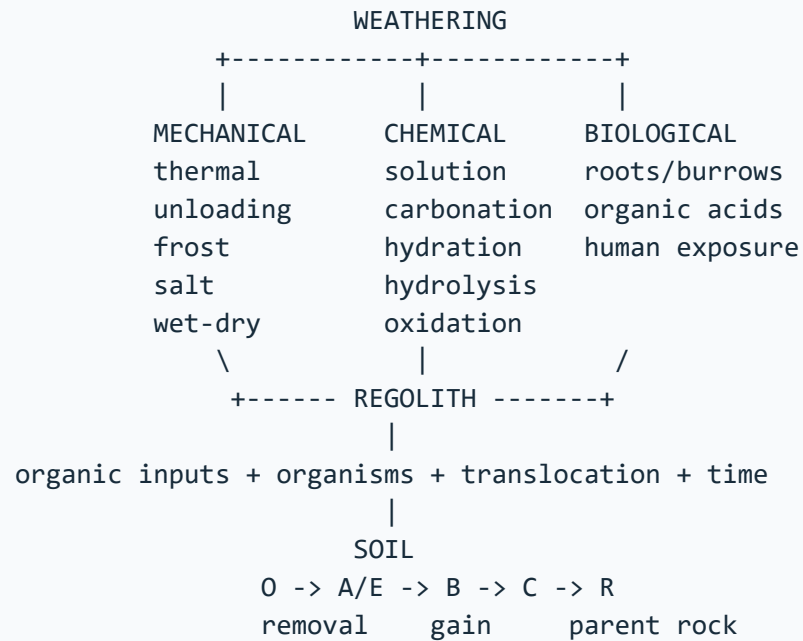


Denudation is collective lowering; weathering, erosion and mass wasting are linked but distinct processes.

MUST REMEMBER: Weathering prepares material in situ, mass movement transfers it downslope, running water removes it, and groundwater follows infiltration, percolation, aquifer storage and discharge; Indian analysis links lithology, slope, rainfall, land use and pumping.

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ASCII MASTER FLOW – PANEL 2/12: Weathering taxonomy, controls, regolith and soil formation



CONTROLS CONVERGE: mineralogy + joints/bedding + climate + relief
+ drainage + organisms + time + land use

Black soil link: Deccan basalt parent, shrink-swell clay, deep dry cracks;
parent material does not alone determine every soil property.

ASCII MASTER FLOW – PANEL 3/12: Soil erosion sequence and watershed control logic

RAINDROP IMPACT + BARE SOIL + RUNOFF CONCENTRATION

```

      |
diffuse sheet removal
      |
small rill channels
      |
entrenched gullies
      |
ravine landscape: Chambal cue
```

WATERSHED RESPONSE

cover soil -> slow runoff -> shorten slope length -> contour/terrace
-> protect drainage -> treat gully head -> rehabilitate catchment

WIND BRANCH

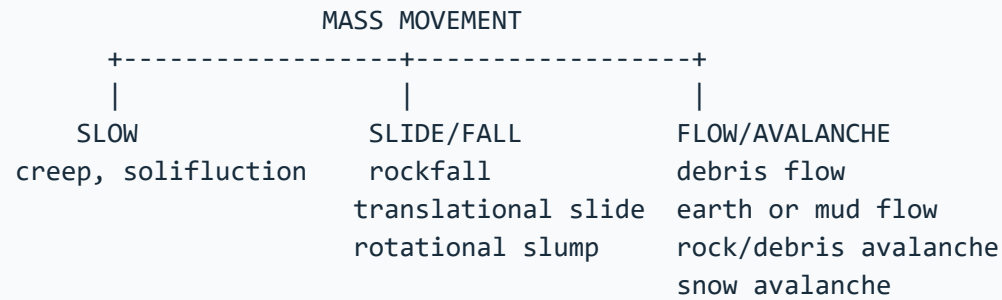
surface creep -> saltation -> suspension

COASTAL BRANCH

wave erosion belongs to the coastal-process system; use only when asked.

A check dam alone cannot replace upstream cover, drainage and maintenance.

ASCII MASTER FLOW – PANEL 4/12: Mass-movement taxonomy and slope-stability mechanism



SLOPE BALANCE

resisting strength = cohesion + friction + structural support

versus

driving stress = gravity component + loading + toe loss

rain infiltration -> added weight + pore pressure -> lower effective strength

road or river cut -> steepening / undercutting -> support loss

quake / quarry / leakage / drainage change -> trigger or amplification

FAILURE WHEN DRIVE > RESISTANCE

Material, movement and water define the event; speed alone does not.

ASCII MASTER FLOW – PANEL 5/12: India landslide geography, risk production and mitigation stack

INDIA LANDSLIDE GEOGRAPHY

Himalaya / NE : young relief + fractures + incision + rain + seismicity
Western Ghats : escarpment + deep weathering + orographic rain + cuts
Nilgiris : weathered slopes + roads + settlements + drainage change

|
PREDISPOSITION -> TRIGGER -> MOVEMENT -> EXPOSURE -> LOSS
geology/slope rain/quake slide/flow road/town disaster

GSI LADDER

|
inventory -> susceptibility/hazard study -> post-event diagnosis
-> monitoring / NLFC support -> decision communication

MITIGATION STACK

|
avoid unsafe siting -> manage drainage -> protect toe
-> stabilise slope -> bioengineer -> control spoil/quarrying
-> maintain works -> warn/close routes -> evacuate and prepare

Inventory maps past events; susceptibility maps relative likelihood;
neither is a deterministic event prediction.
Joshimath is a multi-causal subsidence case, not a synonym for landslide.

ASCII MASTER FLOW – PANEL 6/12: Groundwater architecture from recharge to wells

rain -> infiltration -> soil moisture -> percolation -> recharge

```
GROUND SURFACE      recharge area              spring at slope
-----v----->
ZONE OF AERATION      perched water on local aquitard
----- capillary fringe -----
WATER TABLE          UNCONFINED AQUIFER          pumping well
~~~~~\_____ cone of depression
AQUITARD              slow transmission
-----
CONFINED AQUIFER      pressure; water rises to potentiometric level
=====
AQUICLUDE             practically no transmission
```

porosity = fraction of voids | permeability = connected flow capacity
aquifer stores and transmits | aquitard transmits slowly
artesian water rises under pressure; it flows only if head exceeds ground

Budget: change in storage = recharge + inflow - discharge - pumping - outflow.

ASCII MASTER FLOW – PANEL 7/12: India groundwater depletion, quality and subsidence geography

REGION	AQUIFER FORM	MAJOR PRESSURE
Indo-Gangetic	layered alluvium	depletion, As, nitrate
Peninsular	weathered fractures	discontinuous storage
Deccan basalt	contacts + fractures	uneven well yield
Arid hard rock	limited recharge	fluoride, salinity
Coastal belt	fresh-saline wedge	intrusion and upconing

crop-water demand + cheap energy + private wells + urban growth
+ weak aquifer regulation + limited recharge

falling head -> lower yield -> higher lift cost -> unequal access

compressible sediment: pore-pressure loss -> compaction -> subsidence

Quality paths differ: arsenic redox | fluoride rock-water interaction

nitrate sewage/fertiliser | salinity coastal or inland

Quantity and quality require separate evidence, though each limits usable water.

ASCII MASTER FLOW – PANEL 8/12: Recharge, demand governance and integrated Mains answer spine

GROUNDWATER SECURITY REQUIRES TWO CONVERGING ARMS

SUPPLY / RECHARGE

roof and runoff capture

managed aquifer recharge

watershed treatment

spring-shed restoration

quality-safe source water

maintenance and monitoring

DEMAND / GOVERNANCE

crop-water alignment

efficient irrigation

well registration/metering

conjunctive use and reuse

community water budgets

enforceable state/local rules

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/
+---- AQUIFER PLAN ----+

```

CGWB assessment + NAQUIM mapping + state regulation

+ Atal Bhujal learning + local institutions + public data

ANSWER SPINE

1 define stock-flow-quality system | 2 draw aquifer or India map

3 diagnose regional mechanism | 4 use dated evidence carefully

5 match intervention to cause | 6 qualify limits and equity

Recharge structures are inputs, not proof of water-table recovery.

Desalination supplies water but does not restore freshwater aquifer head.

EVIDENCE LADDER

1. Start with macro-phase, then regional sequence, dated stratum and assemblage.
2. Read cores, flakes, debitage, wear and raw material before inferring behaviour.
3. Separate lithic technology from hominin taxonomy, language and group identity.

VERDICT: Broad chronology guides comparison; site context controls the claim.

ASCII MASTER FLOW – PANEL 10/12: Examiner traps and contested boundaries

CLOSE DISTINCTIONS

1. Acheulian is a technology, Levallois a method and microlith a tool element.
2. Bhimbetka spans many periods; rock-art meaning and dates remain indirect.
3. Mesolithic domestication and sedentariness must be established site by site.

VERDICT: Transition means changing combinations, overlap and regional coexistence.

CLOSE DISTINCTION: Weathering differs from erosion, porosity from permeability, water table from piezometric surface, aquifer from aquitard, and slump/slide/flow/creep require movement-specific geometry rather than the generic label landslide.

ASCII MASTER FLOW – PANEL 11/12: Integrated answer spine and qualified conclusion

ANSWER SPINE

1. Use the 2019 Denisovan PYQ to separate population, place and geological period.
2. Compare phases through tools, ecology, mobility, habitation, art and burials.
3. Write claim -> site/assemblage -> significance -> date/context limit -> verdict.

VERDICT: Indian prehistory is regionally varied history reconstructed without texts.

EVIDENCE LIMIT: A trigger such as intense rain is not the complete cause: antecedent moisture, discontinuities, toe cutting, drainage and construction condition failure; groundwater trends and notified areas require dated official status.

EVIDENCE LADDER

1. = from source book = inference / standard knowledge = current affairs.
2. Agriculture.md uses "chernozem", "alluvial" and "black soil" as...
3. Evidence units available in this file

VERDICT: Source type controls the strength and limits of the historical claim.